

Name: _____ Date: _____

Period: _____

1. **Model from Restrictions.** A farmer has 120 feet of fencing to enclose a rectangular garden. Both pairs of opposite sides require fencing (no wall). Let w = width in feet.

- (a) Write the constraint equation and solve for length ℓ in terms of w .

$$\text{Constraint: } 2w + 2\ell = 120 \Rightarrow \ell = 60 - w$$

- (b) Write the area function $A(w)$.

$$A(w) = w(60 - w) = -w^2 + 60w$$

- (c) State the domain restriction.

$$\text{Domain: } 0 < w < 60$$

- (d) Find the width and length that maximize area.

$$w_{\max} = 30 \text{ ft} \quad \ell_{\max} = 30 \text{ ft} \quad A_{\max} = 900 \text{ ft}^2$$

Teacher Note

Problem 1: the unconstrained garden (no wall) produces a square at maximum area. This contrasts with Lesson 1.14's barn example where the optimal is not square. Good discussion point.

2. **Rational Inverse-Proportion Model.** The gravitational force between two objects is modeled by $F(d) = \frac{k}{d^2}$, where d is the distance in meters and $k = 4 \times 10^{14} \text{ N} \cdot \text{m}^2$.

- (a) State the domain restriction and describe what happens to F as $d \rightarrow \infty$.

$$\text{Domain: } d > 0 \quad \text{As } d \rightarrow \infty, F(d) \rightarrow 0.$$

- (b) Compute $F(2 \times 10^7)$ and $F(4 \times 10^7)$. What is the ratio $F(2 \times 10^7) / F(4 \times 10^7)$?

$$F(2 \times 10^7) = \frac{4 \times 10^{14}}{4 \times 10^{14}} = 1 \text{ N} \quad F(4 \times 10^7) = \frac{4 \times 10^{14}}{16 \times 10^{14}} = 0.25 \text{ N}$$

Ratio: 4 : 1 This illustrates the **inverse-square** law.

3. **Apply the Model — Average Rate of Change.** A ball's position (in feet) after t seconds is modeled by $s(t) = -16t^2 + 64t + 5$.

- (a) Evaluate $s(1)$ and $s(3)$.

$$s(1) = -16 + 64 + 5 = 53 \text{ ft} \quad s(3) = -144 + 192 + 5 = 53 \text{ ft}$$

- (b) Compute the average rate of change of s on $[1, 3]$. Include units.

$$\text{ARC} = \frac{s(3) - s(1)}{3 - 1} = \frac{0}{2} = 0$$

Units: **feet per second (ft/s)**

- (c) Interpret the ARC: what does it tell you about the ball's motion?

An ARC of 0 ft/s means the ball is at the same height at $t = 1$ and $t = 3$ s; the ball went up and came back down symmetrically over this interval (the vertex is at $t = 2$ s)......

4. **Interpret with Units.** A model gives the volume of water $V(t)$ in gallons in a tank t minutes after draining begins. The average rate of change on $[0, 10]$ is -15 gal/min.

(a) Is the tank filling or draining? How do you know from the sign of the ARC?

The tank is draining. The ARC is negative, meaning volume decreases on average as time increases......

(b) If $V(0) = 200$ gal, estimate $V(10)$ using the average rate of change. Is this an exact value or an approximation? Why?

$V(10) \approx 50$ gal. This is an **approximation** because **the ARC is an average; the actual rate of draining may not be constant, so $V(10)$ could differ.**.....

5. **Piecewise Model.** A downtown parking garage charges:

- A flat fee of \$3.00 for the first hour (or any part thereof).
- \$2.00 per hour for each additional hour after the first.

Let $C(t)$ = total cost in dollars for t hours of parking, $t > 0$.

(a) Write the piecewise model. (Treat hours as a continuous variable.)

$$C(t) = \begin{cases} 3 & \text{if } 0 < t \leq 1 \\ 3 + 2(t - 1) & \text{if } t > 1 \end{cases}$$

(b) Compute $C(0.5)$, $C(1)$, $C(2.5)$, and $C(4)$.

$$C(0.5) = \$3.00 \quad C(1) = \$3.00 \quad C(2.5) = \$6.00 \quad C(4) = \$9.00$$

(c) Is $C(t)$ a polynomial function? Explain.

No; $C(t)$ is a piecewise function defined by different rules on different intervals, not a single polynomial expression valid for all $t > 0$

6. **AP-Style Multiple Choice.** A function model for a company's revenue (in thousands of dollars) is $R(x) = -x^2 + 8x$, where x is the number of units sold (in hundreds), $0 \leq x \leq 8$. What is the average rate of change of R on $[2, 6]$?

(A) 0 **← correct** (B) 4 (C) -4 (D) 8

Answer: **A** Show work:

$R(2) = -4 + 16 = 12$; $R(6) = -36 + 48 = 12$; $\text{ARC} = (12 - 12)/(6 - 2) = 0$. The vertex is at $x = 4$, so the function is symmetric about $x = 4$ on $[2, 6]$

Reflection

What strategy do you use first when deciding whether a model should be polynomial or rational? **Check whether the context describes a relationship where one quantity grows by powers (polynomial) or decreases as the other increases according to a reciprocal power (rational/inverse proportion).**